

$$A = \begin{pmatrix} 0.4 & 0.7 \\ 0.6 & 0.3 \end{pmatrix}$$

$$\text{Convention} = \begin{pmatrix} \text{Rain} \\ \text{Dry} \end{pmatrix}$$

$$\begin{aligned} \text{Column} &= \text{From} \\ \text{Row} &= \text{to} \end{aligned}$$

$$\begin{aligned} \text{Chance Rain} &\rightarrow \text{Dry} = 0.6 \\ &\rightarrow \text{Rain} = 0.4 \end{aligned}$$

$$\begin{aligned} \text{Dry} &\rightarrow \text{Dry} = 0.3 \\ &\rightarrow \text{Rain} = 0.7 \end{aligned}$$

Columns add to 1

$\Rightarrow$ Largest eigenvalue 1

$$Av = \lambda v$$

$$\lambda = 1$$

$$Av = v \quad \text{so } v \text{ is equilibrium}$$

v is not changing

100

100

100

100

100

100

100

100

100

100

100

100

100

100

100

100

100

100

100